

A Conditioned Metatheorem for RA-Linearizability of State-Dependent MRDTs

with the tombstone-free RGA as its instance — pen-and-paper proof and mechanization
map

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Abstract

We give a metatheorem: any Mergeable Replicated Data Type whose operations satisfy a fixed set of *conditioned* verification conditions — commutation, a threadable reorder invariant, and a merge-linearization bridge, each *conditioned on a state invariant* `Inv` and an *applicability predicate* `app` — is RA-linearizable, once and for all. The conditioning is what admits *state-dependent* datatypes, whose operations commute and merge correctly only on the reachable, sensible states, and which therefore fall outside the unconditioned metatheory. Our flagship instance is the tombstone-free RGA, a replicated list in which deletion physically removes an element and re-anchors its dependents; it discharges every conditioned VC, the hard ones resting on a single invariant (`RecPathFaithful` below) that is true by construction. The nine flat production MRDTs discharge the same VCs trivially and inherit RA-linearizability nearly for free. Strong eventual consistency is a corollary. Each step carries its Lean name and kernel status.

1 RA-linearizability, and why it must be conditioned

MRDTs. A datatype fixes a state space, an initial state σ_0 , an operation set, a transition function $\text{do} : \text{State} \rightarrow \text{Op} \rightarrow \text{State}$, and a three-way merge $\text{merge} : \text{State}^3 \rightarrow \text{State}$ (arguments: the lowest common ancestor and the two branches). Replicas apply local operations with `do` and reconcile divergent states with `merge`. Each replica state is a *version*; the versions form a DAG whose leaves are updates (`do`) and whose internal nodes are merges.

Linearization and canonical states. Fix an event set E with a visibility (causal) order. The *RA order* lo^E orders the pairs a correct sequential execution must not invert: a visible non-commuting pair by visibility, and a concurrent non-commuting pair by the return-value-aware tie-break (the absorber/overwriter clause — ordered by `rc` unless the later event is already overwritten by a still-later non-commuting one). Two points matter for the proof and were confirmed against the mechanization (`RA_Linearizability.lean`). First, lo^E is *not transitive*: “ π respects lo^E ” is therefore the elementwise condition $\pi.\text{Pairwise}(\lambda a b. \neg \text{lo}^E b a)$ (**respects**), not sortedness, and lo^E is in general partial — many enumerations respect it, so the canonical state $\sigma^*(E) := \text{fold } \sigma_0 \pi$ is well defined only up to $\approx$ and only once such folds are shown to converge (Lemma 1). Second, for the *conditioned* metatheorem lo^E is taken over conditioned commutation `commutesOn` (`MRDTSig.lean`); the unconditioned lo^E over `commutes` (`RA_Linearizability.lean`) is the flat specialization.

Observational equivalence. $\sigma \approx \sigma'$ iff no client operation can distinguish them (equal liveness, payloads, and observable order per element). Representation junk — dead-node residue, internal ordering artefacts — is quotiented away. Convergence and RA-linearizability are always stated up to $\approx$.

Definition 1 (RA-linearizability, per version, up to $\approx$). An MRDT is *RA-linearizable* if, in every well-formed execution, each version v with head state s and delivered event set E_v admits a witnessing linearization: there is a permutation π of E_v (`listPermOf`) that respects lo^E (`respects`) with

$$\text{fold } \sigma_0 \pi \approx s.$$

This is the mechanized `IsRALinearizable` (`RA_Linearizability.lean`) with one change: structural equality $\text{fold } \sigma_0 \pi = s$ is relaxed to observational $\approx$. The relaxation is exactly what a state-dependent datatype needs — the tombstone-free RGA converges only up to observable behavior (dead-node residue may differ), so equality on the nose is too strong. Given convergence (Lemma 1) the witness form is equivalent to $s \approx \sigma^*(E_v)$; we use the existential form as the definition and the canonical form as a convenience.

Why conditioning is forced. The unconditioned metatheory (below) proves RA-linearizability from VCs that quantify over *all* states. That works for flat datatypes — counters, OR-sets, MVR — whose operations commute and merge on every state. It fails for *state-dependent* datatypes. In the tombstone-free RGA, two concurrent inserts commute only when each one’s anchor is faithfully resolvable in the other’s presence; an insert under a since-deleted, physically-removed node is not even applicable. Written over all states, the commutation VC is simply false (a mechanized counterexample: `naive_general_swap_false`). The remedy is to condition every VC on a state invariant `Inv` (what a reachable state guarantees) and an applicability predicate `app` (when an operation is sensible) — and to prove RA-linearizability from the conditioned VCs. The developer’s burden is only to describe sensible operations; the metatheorem does the rest.

2 The conditioned verification conditions

A datatype presents a signature $\langle \text{State}, \sigma_0, \text{do}, \text{merge}, \text{Inv}, \text{app} \rangle$ where `Inv` is a predicate on states and `app` a predicate on (operation, state) pairs (`Lean: ConditionedMRDTSig, MRDTSig.lean`). It must supply:

VC 1 (Discipline). `Inv` σ_0 holds; `do` preserves `Inv` on applicable operations; and generation is disciplined: every operation is `app` and *fresh* at its birth state, with globally unique, monotone ids. (This is the execution model, `Lean: ConditionedConfiguration, ConditionedExecutionModel.lean` — kernel-clean, axiom-free.)

VC 2 (Conditioned commutation — the swap witness). For concurrent operations a, b and any state σ with `Inv` σ at which both are applicable in the relevant reorder sense, $\text{do}(\text{do } \sigma a) b \approx \text{do}(\text{do } \sigma b) a$.

VC 3 (Threadable reorder invariant). An auxiliary invariant J on (state, pending-event) that (i) certifies applicability/the swap precondition of VC 2 at reorder states, (ii) holds at the enabling fold of each event, and (iii) is preserved by each reorder step. Only the *enabled* events — those whose causal past is folded — need be certified; that scope is exactly what a linearization repair ever transposes.

VC 4 (Merge–linearization bridge — the conditioned Join Lemma). For a reachable LCA state l and branches a, b obtained from l by disjoint concurrent event lists, $\text{merge } l a b \approx \text{fold } l \pi$ for a lo^E -respecting enumeration π of the branch events.

For flat datatypes $\text{Inv} = \text{app} = \text{true}$, VC 2 degenerates to unconditioned commutation, VC 3 to $J = \text{true}$, and VC 4 to the ordinary Join Lemma — recovering the unconditioned methodology.

3 The metatheorem

Theorem 1 (Soundness — conditioned VCs $\Rightarrow$ RA-linearizability). *Any MRDT satisfying VCs 1–4 is RA-linearizable (Definition 1).*

The proof is generic — it never inspects a particular datatype — and factors through two lemmas, each proved over the abstract signature with $\approx$ an arbitrary congruence for `do` and `merge`.

Lemma 1 (Convergence). *VCs 1, 2, 3 imply that all lo^E -respecting admissible enumerations of a backward-closed reachable E fold from σ_0 to $\approx$ -equal states; hence $\sigma^*(E)$ is well defined up to $\approx$.*

Proof. Any two lo^E -respecting enumerations of E are related by repeatedly transposing adjacent lo^E -incomparable events. Because lo^E is non-transitive, “respecting” is the elementwise **Pairwise** condition rather than sortedness, so this order-repair is more delicate than for a total order — it is exactly what the generic engine `eq_convergence` carries out, and we appeal to it rather than re-derive it. Each transposition preserves the fold up to $\approx$ by VC 2, provided the swap precondition holds at the transposition point; VC 3 supplies it along the whole enumeration (base at each event’s enabling fold, preserved by each step, scoped to the enabled events — the only ones the repair transposes). The engine consumes only that $\approx$ is an equivalence, `do` is $\approx$ -congruent, and a swap oracle of the shape VC 2+3 provides (Lean: `eq_convergence`, kernel-clean). $\square$

Lemma 2 (Canonical adequacy). *VC 4 and Lemma 1 imply that every reachable version state is $\approx \sigma^*$ of its delivered event set.*

Proof. Induction over the version DAG. Initial: the empty fold. Update node: appends one delivered operation to a canonical fold (definitional). Merge node with LCA l and branches a, b : by hypothesis $a \approx \sigma^*(E_a)$ and $b \approx \sigma^*(E_b)$, and $l \approx \sigma^*(E_l)$. First rewrite a, b to literal canonical folds using $\approx$ -congruence of `merge` (so VC 4, stated for branch folds, applies); then VC 4 gives $\text{merge } l a b \approx \text{fold } l \pi$ for a lo^E -respecting π of the branch events $E_a \cup E_b$. Now E_l causally precedes those branch events (they were generated on states derived from l), so a lo^E -respecting enumeration of $E_l \cup E_a \cup E_b$ factors as (an lo^E -enumeration of E_l) followed by π ; hence $\text{fold } l \pi = \text{fold } (\sigma^*(E_l)) \pi \approx \sigma^*(E_l \cup E_a \cup E_b)$ by Lemma 1 (the engine is start-state generic). Backward closure keeps every enumeration admissible. $\square$

Proof of Theorem 1. Immediate from Lemma 2: each version state $\approx \sigma^*(E_v)$, which is Definition 1. (Unconditioned template: `ra_linearizable3_of_joinC` plus the closure-indexed adequacy bridge, already kernel-clean with nine instances; the conditioned metatheorem generalizes it by carrying `Inv/app` through the two lemmas.) $\square$

Corollary 1 (Strong eventual consistency). *Two versions with the same delivered set E have $\approx$ states: both $\approx \sigma^*(E)$ by Theorem 1, hence $\approx$ each other.*

4 The tombstone-free RGA discharges the conditioned VCs

The RGA is a forest of identified nodes; `do` inserts a fresh node under a resolved anchor or physically deletes a node and rehomes its children; `merge` keeps OR-set survivors and re-anchors each by climbing to the nearest surviving ancestor. `resolve` σp returns the first live id on a recorded path p ; `anc` σx is x 's current parent. Take $\text{Inv} := \text{wf} \wedge \text{id_mono} \wedge (\text{root not reinserted})$ and $\text{app} := \text{accurate} \wedge \text{fresh}$. We discharge each VC; the two hard ones reduce to one invariant.

Definition 2 (Recorded-path faithfulness `RecPathFaithful`). `RecPathFaithful` $a \sigma$: the recorded path `rec` a is a *live subchain of genuine ancestors* of a 's target in σ — true `anc`-ancestors, nearest first, consecutive pairs real parent edges, and exactly the chain live at a 's generation. (This is `accurate/HistFaithful` promoted to a reachability invariant.)

Lemma 3 (`RecPathFaithful` is a reachability invariant). *`RecPathFaithful` $a \sigma$ holds at every reachable σ in which a is applied.*

Proof. Induction on the operations building σ . *Birth:* `do` sets `rec` $a = \text{resolve-path}$ at generation = the live-ancestor chain, so `RecPathFaithful` holds (`chainFaithful_of_histFaithful`). *Later* `Ins` with fresh id f : attaches f as a descendant — no reparenting, no liveness change for prior events' chains. f cannot appear spuriously in any `rec` a : its members are older ids $< a.\text{id} \leq$ every id extant at a 's birth, whereas f exceeds all of them, so $f \notin \text{rec } a$ definitionally. The resurrection/clash regime (`chainFaithful_not_preserved_under_clash_ins`) is therefore *unreachable* by freshness (`fresh_not_mem_recList`, `clash_recList_excluded`), not excluded by a side condition. *Later* `Del` x : flips x 's liveness only; `rec` a 's entries stay genuine ancestors, one possibly now dead — still a live subchain (`chainFaithful_doDel_faithful`). $\square$

Lemma 4 (Key Lemma: resolution over a live subchain). *For any live subchain of genuine ancestors pre of x , in every reachable σ , `resolve` $\sigma pre = \text{anc } \sigma x$. (Lean: `subchain_resolve`, `RGA_Subchain-Resolve.lean`, `kernel-clean`.)*

Proof. `resolve` returns the first entry live in σ ; each entry is a genuine ancestor, so this is the nearest live ancestor among pre . It is the nearest overall: no live ancestor $c \notin pre$ is nearer than that first-live entry — (a) if c was live at capture it was in the captured live chain, so $c \in pre$ (and any pre -entry nearer than the first-live one is dead in σ , hence not c); (b) if c was dead at capture it is still dead (physical, tombstone-free deletion is permanent); (c) an event inserted after capture attaches as a descendant, never a new ancestor of the pre-existing x . The mechanization absorbs these three cases into one preserved invariant `LiveChain` $\sigma x pre := (\text{root unstored}) \wedge \text{live } \sigma x \wedge \text{lsAncPath } \sigma x (pre.\text{filter live})$, which holds at capture, is preserved by each insert (`isAncPath_upd`) and delete (`isAncPath_surgery`, chain splicing across a deleted entry), and yields the conclusion. Case (b) is discharged by an explicit hypothesis $t \notin pre$ on inserts (`okStep`) — the mechanized form of “no resurrection,” itself a consequence of monotone allocation, every pre -entry sitting below x 's id. This replaces the earlier full-chain lemma `hres_of_lchain`, which failed on the proper subchains rehoming produces. $\square$

Proposition 1 (RGA discharge). *The RGA satisfies VCs 1–4.*

Proof. VC 1 (discipline): `Inv` preservation and the generation model are `ConditionedConfiguration` instantiated at the RGA, with the RGA's `do`-invariance lemmas; `conditioned_premises` supplies the timestamp/freshness facts. VC 2 (swap): `general_swap_bothFaithful` / `eqSwap_of_bothFaithful` — concurrent operations commute up to $\approx$ when both are *faithful* (neither

need be exact/accurate), kernel-clean. *VC 3* (reorder invariant): $J := \text{ChainFaithful}$ on the enabled scope, with steps `chainFaithful_incompStep/incompFold`; its *base* — each event faithful at its enabling fold — is exactly `RecPathFaithful` (Lemma 3) projected through the Key Lemma (Lemma 4): the residual goal $\text{resolve } \sigma((\text{rec } a) \setminus t') = \text{anch}'$ when an ancestor t' becomes leftmost live is the Key Lemma on the remaining subchain. *VC 4* (Join): per-id extensional match (`eq_merge2_of_branchInv2`); the anchor-coincidence clause is the single-sided I_4 (`eq_merge_single`, kernel-clean) and, cross-branch, the Key Lemma directly (`branchInv_doDel_crossBranch`, kernel-clean); order-independence is Lemma 1 at start state l . Both hard VCs (3 base, 4 cross-branch) thus rest on `RecPathFaithful` and the Key Lemma — proved by construction from freshness and permanence. $\square$

Corollary 2. *The tombstone-free RGA is RA-linearizable (Theorem 1 + Proposition 1), and strongly eventually consistent (Corollary 1). To our knowledge this is the first mechanized RA-linearizability / SEC result for an RGA with physical, tombstone-free deletion (cf. Gomes et al., OOPSLA'17, for the tombstoned RGA).*

Remark 1 (Other instances). The nine flat production MRDTs (counter, OR-set, MVR, ...) take `Inv = app = true`: VC 2 is their unconditioned commutation, VC 3 is $J = \text{true}$, and VC 4 is the ordinary Join Lemma — already discharged (`ra_linearizable3_of_joinC`). They inherit RA-linearizability from the same Theorem 1. The RGA is the instance that exercises every conditioned mechanism; the framework supplies the generality, the RGA the novelty.

5 Mechanization status

Kernel-clean means axioms $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$, no `sorryAx/native_decide/ofReduceBool`.

Component	Lean	Status
Unconditioned metatheorem (template)	<code>ra_linearizable3_of_joinC</code> + adequacy	proved (9 instances)
Convergence engine (Lemma 1)	<code>eq_convergence</code>	proved
Execution model (VC 1)	<code>ConditionedConfiguration</code> , <code>conditioned_premises</code>	proved (axiom-free)
Swap / VC 2	<code>general_swap_bothFaithful</code>	proved
Reorder steps (VC 3)	<code>chainFaithful_incomp-</code> <code>Step/incompFold</code>	proved
Faithful delete	<code>chainFaithful_doDel_faith-</code> <code>ful</code>	proved
Merge single-sided (I_4)	<code>eq_merge_single</code>	proved
Merge cross-branch	<code>branchInv_doDel_cross-</code> <code>Branch</code>	proved
Two-sided merge (VC 4)	<code>eq_merge_two_sided</code>	mod. <code>RecPathFaithful</code>
Key Lemma (4)	<code>subchain_resolve</code>	proved
RecPathFaithful invariant (3)	<code>RGA_RecPathFaithful</code>	in mechanization
RGA discharge (Prop. 1)	<code>RGA_update_convergence</code> + <code>merge</code>	awaits <code>RecPathFaithful</code>
Generic Join $\Rightarrow$ RA-lin (Inv-conditioned, structural =)	<code>ra_linearizable3_of_joinC</code>	proved (tem- plate)
$\approx$ -quotient state layer (hosts the RGA on the =-bridge)	(M5; no quotient exists yet)	to build
app-conditioning on top of Inv	(extends the template)	to build

Reading. Be precise about what exists. The *Inv-conditioned* Join $\Rightarrow$ RA-linearizability theorem is already mechanized and kernel-clean (`ra_linearizable3_of_joinC`: it concludes RA-linearizability from the closure-indexed ternary Join Lemma and already carries `Inv`), with nine flat instances. So Theorem 1 is not built from nothing; two *genuinely new* pieces extend that template, and — verified against `Adequacy.lean` — neither is free transcription. (i) The template is stated over *structural* equality (`IsRALinearizable3` demands $\text{fold } \sigma_0 \pi = s$, and its Join Lemma is $s = \text{merge } lt(\text{update})$), whereas the RGA converges only up to $\approx$. Reconciling these is a real construction, and precisely the deferred M5 layer: either re-prove the bridge over an $\approx$ -congruence (a substantial re-proof of a currently kernel-clean file), or — cleaner, and what we adopt — present the RGA state as a quotient by $\approx$, so structural = on the quotient *is* $\approx$ and the existing =-bridge hosts the RGA unchanged; the cost is lifting `do` and `merge` to the quotient, which they permit since they are $\approx$ -congruent (`do_eq_congr`, `applySeqR_eq_congr`). No such quotient layer exists yet. (ii) Adding the applicability conditioning `app` on top of `Inv` extends the template’s premises. The two lemmas of Theorem 1 — the convergence engine and the adequacy induction — are then reused: the engine already over $\approx$, the adequacy bridge over the quotient. On the RGA side, the discharge is kernel-clean (swap, single-sided and cross-branch merge, reorder steps) except for the two facts it all reduces to: the Key Lemma (now *proved*, `subchain_resolve`) and the `RecPathFaithful` invariant (in mechanization, `RGA_RecPathFaithful`) — both established on paper here (Section 4) by construction. Property-based testing corroborates every RGA step (600 merge-vs-fold triples through the hard corners, zero counterexamples; update-layer threading green across hundreds of interleavings), so no open mathematical question remains. The residual is genuine but bounded:

transcription of `RecPathFaithful`; the $\approx$ -quotient state layer (M5) and the `app` premise-extension of the template — real constructions, not transcription, as the check against `Adequacy.lean` above shows; and the mechanical assembly they unlock.